

In-class activity

STAT 720 – Summer 2026

2026-06-18

A study was designed to determine the effect on the incidence of root rot, of variety of wheat, kinds of dust for seed treatment, method of application of the dust, and efficacy of soil inoculation with the root-rot organism. We have a data frame with 160 observations on the following 9 variables that describe a designed experiment:

Exercises

Data structure

1. Draw a figure of the structure in the data. Do not consider the treatments yet.

Designed experiment

- 2a. Draw a figure of the different randomization steps.

- 2b. Indicate the experimental units for the different treatment factors.

2c. How many observations do you have for each treatment factor?

2d. What are the treatment structure and the data structure?

Stat model

3a. Write the statistical model that describes the data generating process.

3b. Fit the model in (3) to the data using R. Write down the estimated variance components.

Bonus

4. How would you name this experiment design?